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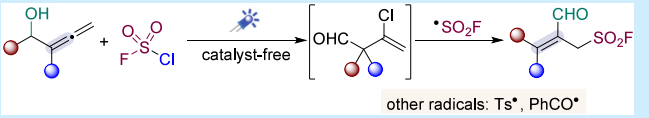
Photoinduced Double 1,2-Carbon Migration toward All-Carbon Tetrasubstituted Olefins

Tianyu He, Chao Kong, Yukai Liu, Yu Zheng, Yanju Lu, Xiaoxiang Zhang, and Shenlin Huang*

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ABSTRACT: A photochemical double 1,2-carbon migration approach was achieved as an efficient entry into all-carbon tetrasubstituted olefins from allenols. Key to the success is the use of FSO_2Cl, which plays a crucial triple role as an effective Cl^+ synthon to facilitate semipinacol rearrangement, an electron acceptor to form electron donor−acceptor complexes, and a $FSO_2^•$ precursor to trigger 1,2-carbonyl migration. The synthetic utility is further showcased by extending it to other radical precursors and diverse follow-up reactions.



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All-carbon tetrasubstituted olefins are important structural motifs that play a key role in pharmaceuticals, natural products, and organic functional materials.[1] Consequently, a variety of classic synthetic strategies to access tetrasubstituted olefins have been developed, including the Wittig reaction, the Horner−Wadsworth−Emmons reaction, Peterson olefination, olefin metathesis, and hydroxy elimination.[2] Over the past few decades, transition-metal-catalyzed cross-couplings have been identified as additional versatile approaches for the construction of these motifs.[2b,c] Recently, radical functionalization of alkynes and alkenes via photochemical methods has offered alternative approaches for the synthesis of tetrasubstituted olefins.[3] In particular, the Xie group demonstrated that metallophotoredox deoxygenative alkenylation of aromatic acids can effectively produce all-carbon tetrasubstituted alkenes (Scheme 1A, top).[4] Very recently, Kleij reported a conceptually novel photoredox method of preparing all-carbon tetrasubstituted alkenes through decarboxylative coupling of vinyl cyclic carbonates and α-amino radicals (Scheme 1A, bottom).[5] Nonetheless, these methods typically require transition metal catalysts, and radical approaches for the stereoselective construction of all-carbon tetrasubstituted olefins still remain rare.

Scheme 1. Context of the Work

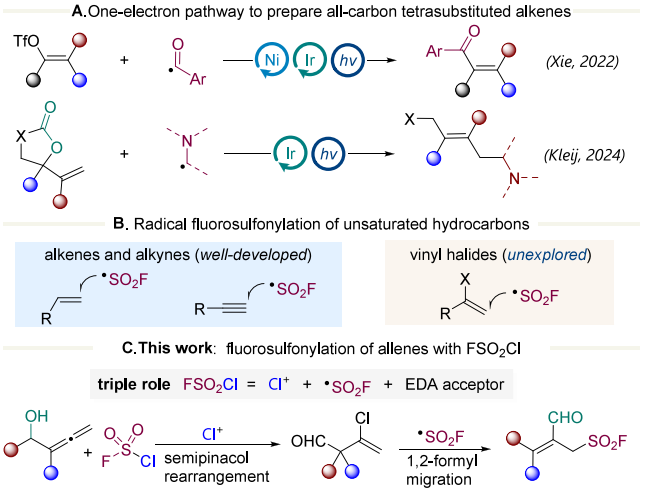
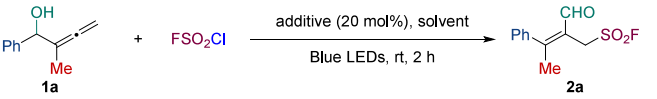


Table 1. Optimization of the Reaction Conditions[a]



entry	additive	solvent (0.05 M)	yield[b]
1	AgOAc	Et_2O	41%
2	Ag_2SO_4	Et_2O	20%
3	$AgNO_2$	Et_2O	43%
4	Ag_2CO_3	Et_2O	62%
5	Ag_2CO_3	THF	0%
6	Ag_2CO_3	2-Me-THF	<10%
7	Ag_2CO_3	CPME or MTBE	0%
8	Ag_2CO_3	1,4-dioxane	50%
9	Ag_2CO_3	1,4-dioxane (0.067 M)	19%
10	Ag_2CO_3	1,4-dioxane (0.04 M)	42%
11	Ag_2CO_3	1,4-dioxane/Et_2O (20/1)	86%
12		**1,4-dioxane/Et_2O(20/1)**	**86%**

[a]Standard conditions: **1a** (0.1 mmol) and FSO_2Cl (0.5 mmol, 5.0 equiv, 1 M in $PhCF_3$), 1,4-dioxane (2 mL), Et_2O (0.1 mL), 35 W blue LEDs, 2 h, rt. CPME = cyclopentyl methyl ether. MTBE = methyl *tert*-butyl ether. [b]Isolated yield.

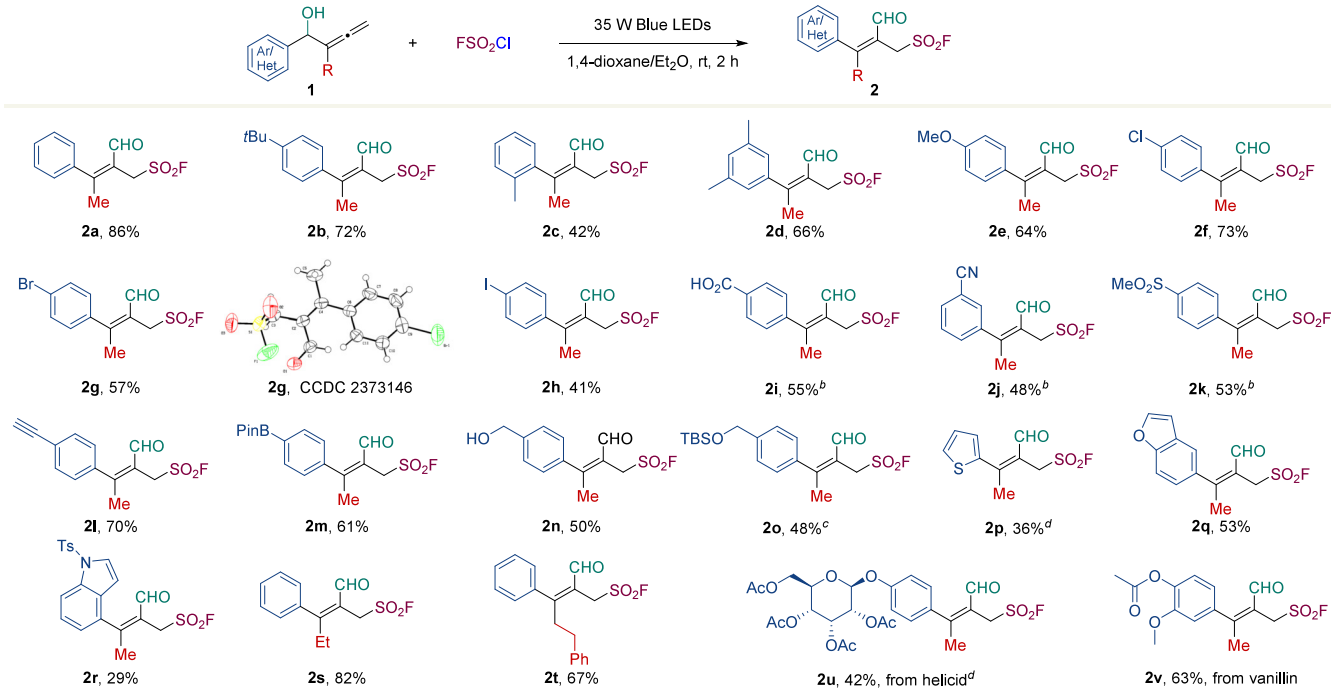
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Scheme 2. Substrate Scope



[a]Conditions: allenes (0.1 mmol) and FSO_2Cl (0.5 mmol, 5.0 equiv, 1 M in $PhCF_3$), 1,4-dioxane (0.05 M), Et_2O (1 M), 35 W blue LEDs, 2 h, rt. Isolated yields. [b]Additional Ag_2CO_3 (1.5 equiv). [c]Additional Na_2CO_3 (2 equiv). [d]NBS was used for semipinacol rearrangement.

In recent years, sulfonyl fluorides have been recognized as privileged building blocks, which undergo sulfur(VI) fluoride exchange (SuFEx) click reactions.[6] Sulfuryl chlorofluoride (FSO_2Cl), a useful solvent, has been successfully used as a dehydrating agent, an oxidant, and a source of F^+, FSO_2^+, or Cl^+.[7] In 2020, Liao disclosed that FSO_2Cl could act as a fluorosulfonyl radical ($FSO_2^\bullet$) precursor for the synthesis of alkenyl sulfonyl fluorides.[8] After that, a few radical fluorosulfonylation strategies were developed using FSO_2Cl under photochemical[9] or electrochemical[10] conditions. Recently, we demonstrated that FSO_2Cl can simultaneously serve as a source of $FSO_2^\bullet$ and as an electron acceptor to form electron donor−acceptor (EDA) complexes in the fluorosulfonylation transformation.[11] However, FSO_2Cl has been mainly used in a monofunctional fashion so far. We envisioned that it would be highly desirable and attractive to make full use of FSO_2Cl as a multifunctional reagent to facilitate continuous reaction sequences.

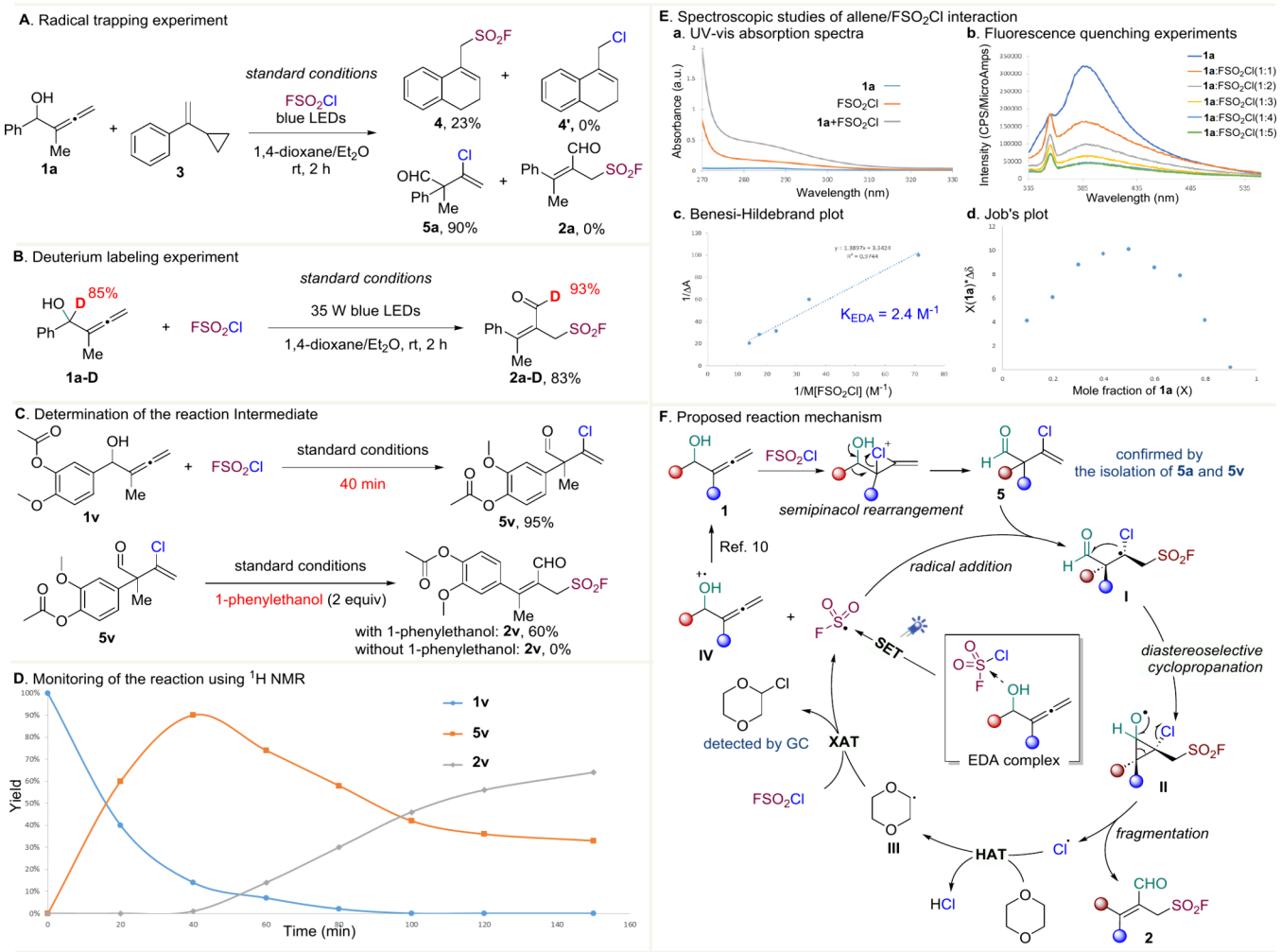
Various synthetic methods have been developed for the introduction of the sulfonyl fluoride group,[12] particularly via radical fluorosulfonylation of alkenes or alkynes,[13] whereas the fluorosulfonylation of vinyl halides remains elusive (Scheme 1B). Herein, we report an unprecedented photochemical method to construct all-carbon tetrasubstituted olefins from allenols with FSO_2Cl through double 1,2-carbon migration (Scheme 1C). In contrast to sulfonyl radicals, the fluorosulfonyl radical adds to the terminal carbon of allenes in our reaction. In this process, FSO_2Cl plays a triple role: an effective Cl^+ synthon to facilitate semipinacol rearrangement to provide the vinyl chloride intermediate,[14] an electron acceptor to form EDA complexes, and a $FSO_2^\bullet$ precursor to initiate radical carbonyl migration.[15] In addition, the generated sulfonyl fluorides possess an aldehyde, a tetrasubstituted alkene, and an allylic sulfonyl fluoride, thus bestowing diverse postsynthetic utilities.

We started off by investigating the reaction of allenol **1a** with FSO_2Cl under blue LED irradiation at ambient temperature for 2 h (Table 1). Originally, we thought silver salts might function as Lewis acids or chloride scavengers to promote the reaction, and several silver salts were examined using diethyl ether as the solvent (entries 1−4). Much to our delight, the FSO_2-functionalized tetrasubstituted olefin **2a** was isolated in 62% yield when the reaction was performed in the presence of Ag_2CO_3 (20 mol %) in diethyl ether (entry 4). Other solvents, such as THF, 2-Me-THF, CPME, and MTBE, delivered the desired product **2a** in 0−10% yields (entries 5−7). Using 1,4-dioxane as the solvent, the desired product **2a** was isolated in 50% yield (entry 8). Increasing or reducing the reaction concentration led to diminished yields (entries 9 and 10). Further optimization revealed that the mixed solvent 1,4-dioxane/Et_2O (20/1) was found to be optimal (entry 11). A control experiment demonstrated that Ag_2CO_3 was unnecessary for this reaction (entry 12). No conversion of **1a** was observed when the 1-fluorosulfonyl benzimidazolium salts developed by Liao[16] and Wang[17] independently were used as FSO_2 radical precursors, further demonstrating the unique role of FSO_2Cl in our reaction.

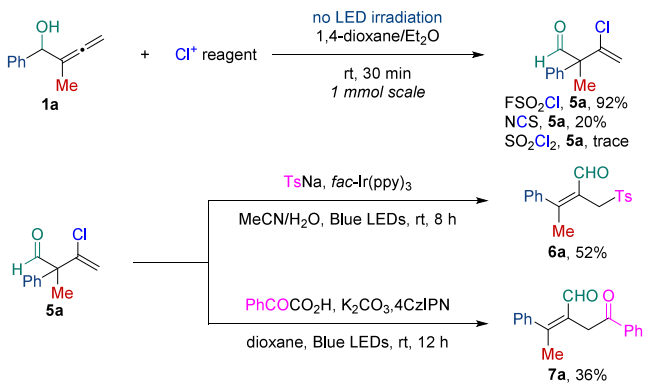
Having identified suitable reaction conditions for tetrasubstituted olefin synthesis, the substrate scope of allenols was investigated (Scheme 2). The aryl ring was initially modulated with regard to electronic properties and the substitution pattern (**2a**−**2o**). Electron-donating groups (*t*Bu, Me, and MeO) were well accommodated, providing desired products **2b**−**2e** in moderate to good yields. Introducing a methyl group at the *ortho* position of the benzene ring led to a notable decrease in yield (**2c**) compared to *para*- and *meta*-substituted derivatives (**2b**, **2d**, and **2e**), presumably due to the steric

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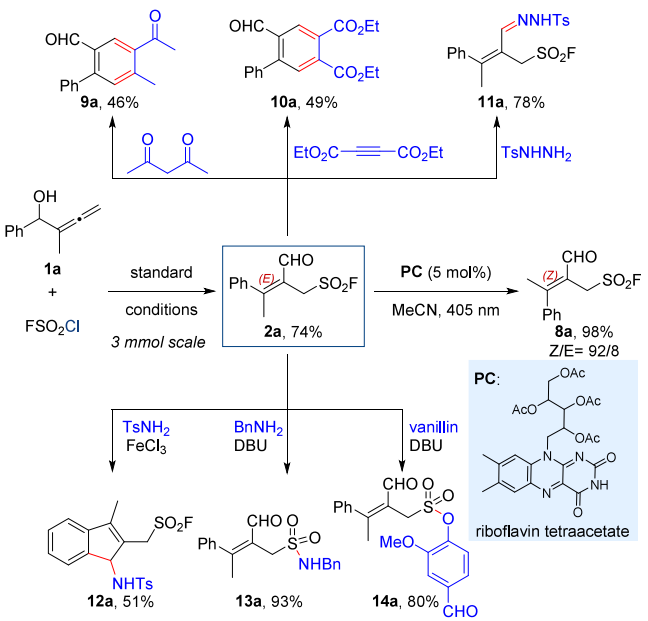
Scheme 3. Mechanistic Investigations and Plausible Reaction Mechanism



Scheme 4. Use of Other Radical Precursors



Scheme 5. Product Derivatizations



effect. Halogens (Cl, Br, and I) and electron-withdrawing groups (CO_2H, CN, and SO_2Me) were tolerated, delivering desired sulfonyl fluorides **2f**–**2k**. Other useful functional groups, including alkyne and boronic ester, were well-tolerated to afford **2l** and **2m**, which are poised for further derivatization. Notably, the reactive benzyl alcohol (**2n**) and silyl ether (**2o**) also remained intact under our reaction conditions. Moreover, allenols bearing thiophene (**2p**), benzofuran (**2q**), and indole (**2r**) heterocycles are competent substrates for this cascade reaction. Subsequently, the presence of a methyl group on the terminal allene carbon was found to be not mandatory, since the method is applicable to ethyl- and phenylethyl-substituted

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allenols (**2s** and **2t**). Delightedly, allenols from the naturally occurring aldehydes helicid and vanillin could be easily converted into sulfonyl fluorides **2u** and **2v**, respectively. In addition, NBS was used to form more reactive vinyl bromide intermediates through semipinacol rearrangement of **1p** and **1u**, thus providing **2p** and **2u** in reasonable yields.

We initiated our mechanistic investigations by conducting a radical trapping experiment (Scheme 3A). When (1-cyclopropylvinyl)benzene **3** was added to the standard reaction of **1a** with FSO_2Cl, dihydronaphthalene **4** was isolated in 23% yield, thereby indicating the involvement of a reaction sequence of fluorosulfonyl radical addition/cyclopropyl ring-opening/cyclization. Meanwhile, **1a** was converted to intermediate **5a** by the reaction with FSO_2Cl via semipinacol rearrangement, and **4′** or **2a** was not obtained. In addition, when **1a-D** was subjected to the standard conditions, **2a-D** was isolated in 83% yield, strongly supporting the formyl migration is involved in the reaction (Scheme 3B). Interestingly, when the reaction of **1v** and FSO_2Cl was quenched after 40 min, vinyl chloride **5v** could be isolated in 95% yield (Scheme 3C). Resubjecting **5v** to the standard conditions with additional 1-phenylethanol as the electron donor produced desired product **2v** in 60% yield, suggesting that **5v** is a plausible reaction intermediate. When the reaction **1v** and FSO_2Cl was monitored by ^1H NMR throughout its entire course, the conversion of allenol **1v** to **5v** and further to sulfonyl fluoride **2v** as the final product was observed clearly (Scheme 3D).

Moreover, several spectroscopic studies were also performed to probe the generation of $FSO_2^{\bullet}$ via EDA photoactivation (Scheme 3E). The ultraviolet−visible (UV−vis) absorption experiments showed a significant red shift for the mixture of **1a** and FSO_2Cl (Scheme 3E(a)). The fluorescence quenching studies showed that allenol **1a** could be efficiently quenched by FSO_2Cl (Scheme 3E(b)). Based on the Benesi−Hildebrand method, an association constant (K_{EDA}) of 2.4 M^{-1} between **1a** and FSO_2Cl in $PhCF_3$ was obtained (Scheme 3E(c)). Additionally, a 1:1 EDA complex between **1a** and FSO_2Cl was demonstrated by Job's plot analysis (Scheme 3E(d)).

Based on these mechanistic studies, a plausible mechanism is outlined in Scheme 3F. Allenol **1** would first undergo semipinacol rearrangement with FSO_2Cl as the effective Cl^+ synthon,[18] resulting in vinyl chloride **5**. Meanwhile, visible light irradiation of the EDA complex, formed by the association of donor **1** with acceptor FSO_2Cl, generated the FSO_2 radical. Then, radical addition to the terminal olefin would occur, affording the alkyl radical **I**. Radical **I** underwent a diastereoselective cyclization to the aldehyde group, leading to alkoxy radical **II**. Subsequent fragmentation of the halogen-substituted cyclopropane[19] **II** provided the desired sulfonyl fluoride **2** along with a chlorine radical. The resulting chlorine radical could then facilitate hydrogen atom transfer (HAT) with 1,4-dioxane to give α-oxo carbon-centered radical **III**, which engaged in halogen atom transfer (XAT) with FSO_2Cl to deliver the FSO_2 radical.

We continued the studies by extending the reaction partners to other sources of radicals using a two-step procedure (Scheme 4). While halogen-promoted semipinacol rearrangement of allenols has been reported by Ma,[18] electrophilic halogenated reagents are restricted to "Br^+" and "I^+". We found that FSO_2Cl could function as an efficient Cl^+ reagent to facilitate the 1,2-aryl migration of allenols. Vinyl chloride **5a** was isolated in 92% yield from the reaction of **1a** with FSO_2Cl. In contrast, the use of NCS and SO_2Cl_2 as halogenated reagents resulted in inefficient rearrangement, leading to the desired product **5a** in 20% yield and a trace, respectively. Both sodium *p*-toluenesulfonate and phenylglyoxylic acid proved to be suitable radical precursors to react with **5a**, furnishing the corresponding tetrasubstituted olefins **6a** and **7a**.

To demonstrate the synthetic utility of our methodology, we performed several synthetic transformations on product **2a** (Scheme 5). Photosensitized alkene isomerization is a powerful synthetic strategy,[20] which offers the opportunity for the stereodivergent synthesis of both *E* and *Z* isomers. Thus, we developed the isomerization conditions using riboflavin tetraacetate as the photosensitizer under 405 nm irradiation and converted the thermodynamically favored *E*-isomer **2a** to less stable *Z*-isomer **8a**. Next, **2a** underwent [4 + 2] annulation with acetylacetone or diethyl acetylenedicarboxylate, leading to tetrasubstituted arenes **9a** or **10a**, respectively. The aldehyde group within **2a** was then condensed with $TsNHNH_2$ to give *N*-Tosylhydrazone **11a**, setting the stage for diverse follow-up functionalization. Treatment of **2a** with $TsNH_2$ in the presence of $FeCl_3$ resulted in aminoindene **12a** through tandem condensation/Friedel−Crafts cyclization.[21] Finally, sulfonamide **13a** and sulfonate **14a** could be obtained in high yields from SuFEx reactions of **2a**.

In conclusion, we have developed an unprecedented photochemical method to construct all-carbon tetrasubstituted olefins from allenols with FSO_2Cl. The operationally simple protocol proceeds via a semipinacol rearrangement/radical 1,2-formyl migration cascade in an excellent regio- and stereoselective manner in the absence of any catalysts and additives. Mechanistic studies point out that FSO_2Cl acts as an effective Cl^+ synthon to facilitate semipinacol rearrangement, an electron acceptor to form EDA complexes, and a $FSO_2^{\bullet}$ precursor to initiate radical 1,2-carbonyl migration. Furthermore, other types of radical precursors can be also applicable to this method, and the multifunctional tetrasubstituted olefins bestow diverse postsynthetic utilities.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c01982.

Experimental procedures, characterization data, mechanistic studies, crystallographic data for **2g** (2373146), and NMR spectra (PDF)

Accession Codes

Deposition Number 2373146 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

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Notes

The authors declare no competing financial interest.

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